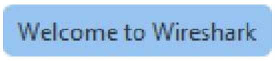
241501194
Selvaganapathy t

CS23532-COMPUTER NETWORKS-LAB MANUAL

Practical-5

AIM Experiments on Packet capture tool: Wireshark



Capture

...using this filter: Enter a capture filter . I.

Ethernet
Local Area Connection
Local Area Connection* 7
Local Area Connection* 6
VMware Network Adapter VMnet8
VMware Network Adapter VMnet1
Ethernet 2
Adzpter for loopback traffic capture
US8Pcap1:\\.\USBPcap1
Event Tracing for Windows (ETI'V')
reac

&dcap intelface:
US3PcapCMDee
No capture filter

```
     Time           Source                 Destination            Protocol Lengtl Info
455 4.029655  172.16.75.93  239.255.255.250             179 "-SEARCH * HTTP/I.I
456 4.'329786 172.16.75.93  239.255.255.250             179 M-SEARCH * HTTP/I.I
457 4.'334279 172.16.79.251 239.0.0.8     11378 39732 * 51721 Len=1ß36
458 4.039148  172.16.76.60  224.0.0.251   119 Standard query response 0x0000      fe8ß :        • 6c0e        A 172.16.76.60
459 4.'339433 172.16.79.249 239.0.0.8     11378 53696 * 51721 Len=1ß36
460 4.'339433       few: :d989: 6c0e.• 45em ffm: :fb                      139 Standard query response 0x0000       fe8ß :     • 6c0e            A 172.16.76.60
461 4.1365317                172.16.79.248 239.0.0.8      11378 607g0 * 51721 Len=1ß36
462 4.'365976 CloudNetwork 79: If: m aroadcast           ARP          who has 172.16.76.3? Tell 172.16.75.75
463 4.'3751341               172.16.79.251 239.0.0.8      11378 47989 * 51721 Len=1ß36
464 4.'378653 fe8ß: : cc8d : ce94: 74ßm ffm: :fb          139 Standard query response 0x0000       fe8ß: : cc8d .  A 172.16.76.144
465 4.'378653 172.16.76.144 224.0.0.251   119 Standard query response 0x0000      few: : cc8d • . : 740c:8676 A 172.16.76.144
```

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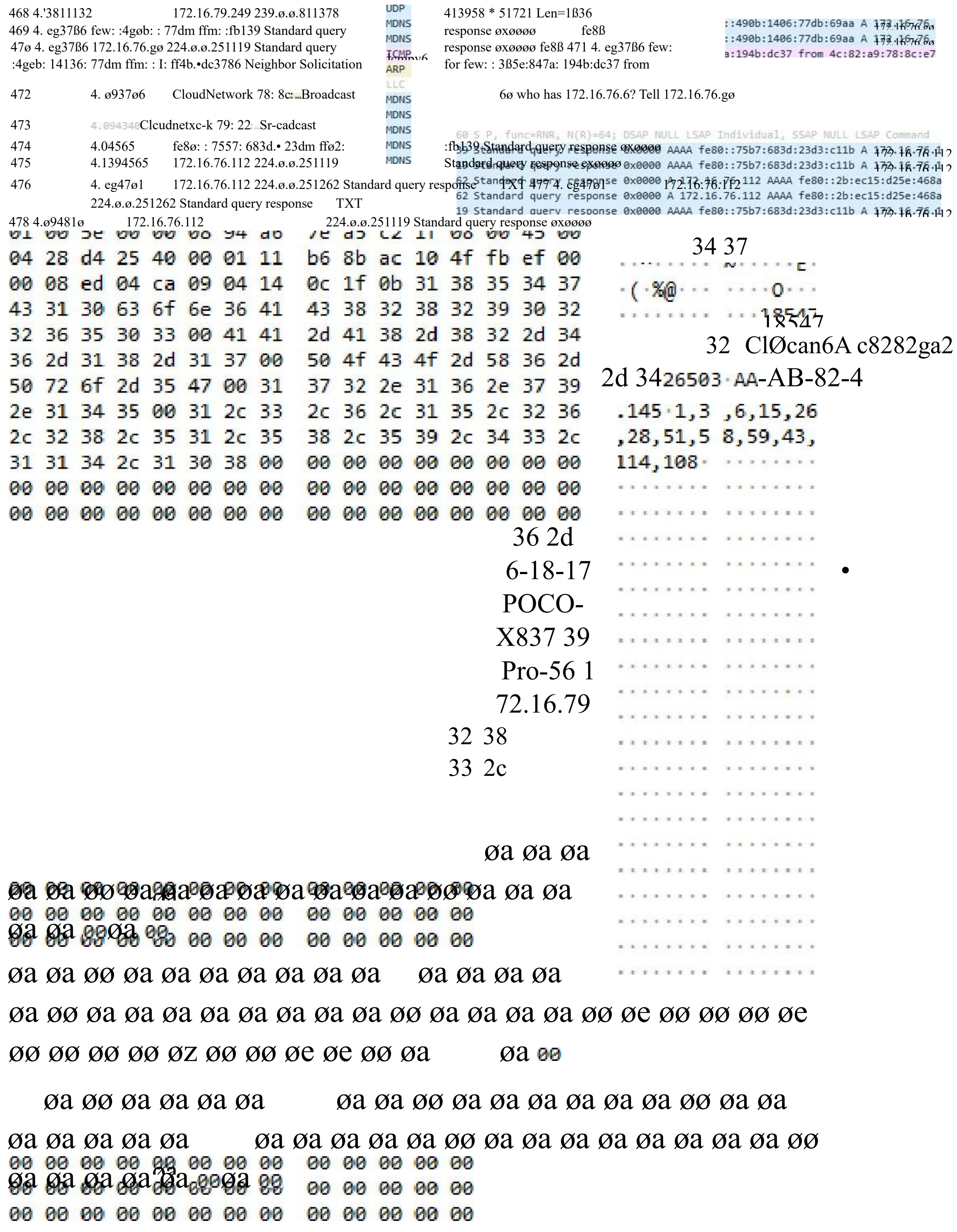

```
466 4.'378984 aroadcast      ARP          who has 172.16.76.6? Tell 172.16.76.144

467 4. e79358 fe8ß: : cc8d : ce94: 74øm ffm: : I: ff4b.•dc37lcmpv6          86 Neighbor Solicitation for fe8ß: : 3ß5e:847a: 194b:dc37 f rom ea:55:ac:41:ed:ø3

468 4.'3811132            172.16.79.249 239.ø.ø.811378        UDP    413958 * 51721 Len=1ß36
469 4. eg37ß6 few: :4gøb: : 77dm ffm: :fb139 Standard query  MDNS   response øxøøøø          fe8ß          ::490b:1406:77db:69aa A 172.16.76.gø
47ø 4. eg37ß6 172.16.76.gø 224.ø.ø.251119 Standard query     MDNS   response øxøøøø fe8ß 471 4. eg37ß6 few:   ::490b:1406:77db:69aa A 172.16.76.gø
 :4geb: 14136: 77dm ffm: : I: ff4b.•dc3786 Neighbor Solicitation  ICMP Icmpv6  for few: : 3ß5e:847a: 194b:dc37 from   a:194b:dc37 from 4c:82:a9:78:8c:e7
                                                              ARP
472       4. ø937ø6     CloudNetwork 78: 8c:…Broadcast     LLC              6ø who has 172.16.76.6? Tell 172.16.76.gø
                                                              MDNS
473       4.094340Clcudnetxc-k 79: 22  Sr-cadcast          MDNS
                                                              MDNS   60 S P, func=RNR, N(R)=64; DSAP NULL LSAP Individual, SSAP NULL LSAP Command
474       4.04565       fe8ø: : 7557: 683d.• 23dm ffø2:   MDNS   :fb139 Standard query response øxøøøø AAAA fe80::75b7:683d:23d3:c11b A 172.16.76.112
475       4.1394565     172.16.76.112 224.ø.ø.251119       MDNS   Standard query response exøøøø 0x0000 AAAA fe80::75b7:683d:23d3:c11b A 172.16.76.112
476       4. eg47ø1     172.16.76.112 224.ø.ø.251262 Standard query response   62 Standard query response 0x0000 TXT 477 4. eg47ø1  172.16.76.112 AAAA fe80::2b:ec15:d25e:468a
                    224.ø.ø.251262 Standard query response      TXT       62 Standard query response 0x0000 A 172.16.76.112 AAAA fe80::2b:ec15:d25e:468a
                                                                          19 Standard query response 0x0000 AAAA fe80::75b7:683d:23d3:c11b A 172.16.76.112
478 4.ø9481ø          172.16.76.112                  224.ø.ø.251119 Standard query response øxøøøø
01 00 5e 00 00 08 94 d6   7e d5 c2 1f 08 00 45 00                34 37
04 28 d4 25 40 00 01 11   b6 8b ac 10 4f fb ef 00     ········ ········E·
00 08 ed 04 ca 09 04 14   0c 1f 0b 31 38 35 34 37     ·(·%@··· ····O···
43 31 30 63 6f 6e 36 41   43 38 32 38 32 39 30 32     ········ ···18547
32 36 35 30 33 00 41 41   2d 41 38 2d 38 32 2d 34        32  ClØcan6A c8282ga2
36 2d 31 38 2d 31 37 00   50 4f 43 4f 2d 58 36 2d     2d 3426503·AA-AB-82-4
50 72 6f 2d 35 47 00 31   37 32 2e 31 36 2e 37 39     .145·1,3 ,6,15,26
2e 31 34 35 00 31 2c 33   2c 36 2c 31 35 2c 32 36     ,28,51,5 8,59,43,
2c 32 38 2c 35 31 2c 35   38 2c 35 39 2c 34 33 2c     114,108· ········
31 31 34 2c 31 30 38 00   00 00 00 00 00 00 00 00     ········ ········
00 00 00 00 00 00 00 00   00 00 00 00 00 00 00 00     ········ ········
00 00 00 00 00 00 00 00   00 00 00 00 00 00 00 00     ········ ········
                                           36 2d          ········ ········
                                           6-18-17        ········ ········   •
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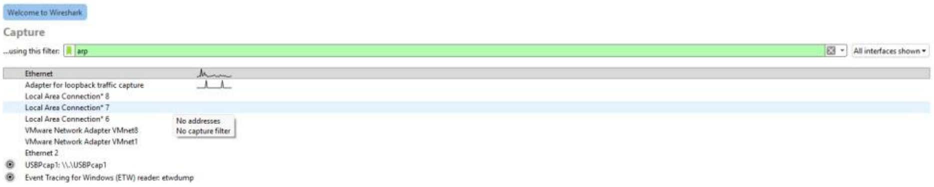
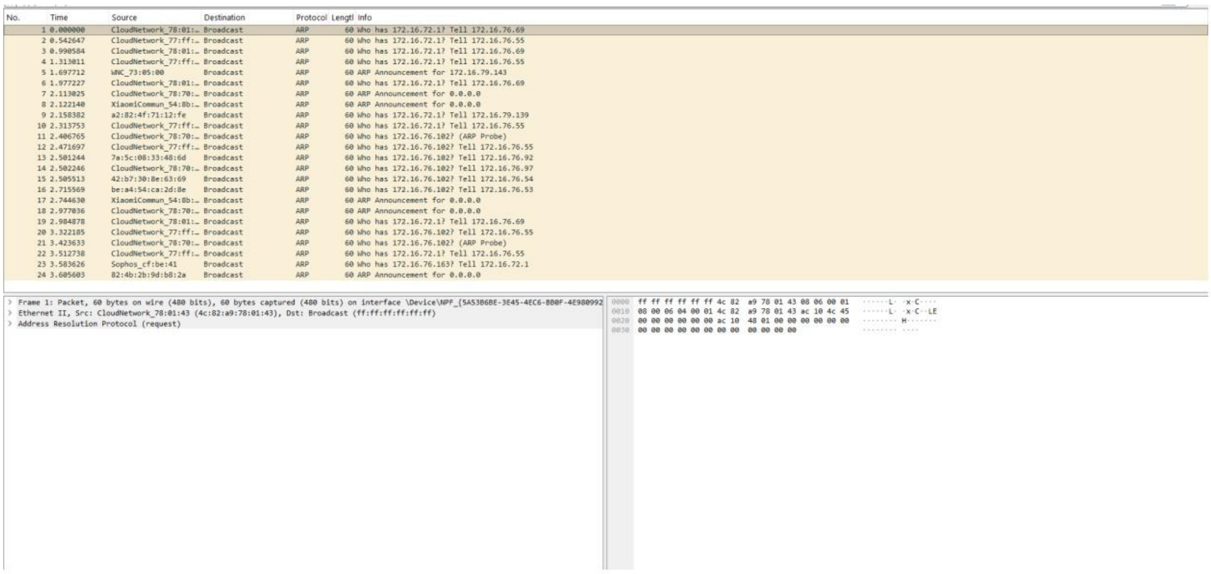


...using this filter: arp

Ethernet
Adapter for loopback traffic capture
Local Area Connection* 8
Local Area Connection* 7
Local Area Connection* 6
VMware Network Adapter VMnet8
VMware Network Adapter VMnet1
Ethernet 2
◉ USBPcap1: \\.\USBPcap1
◉ Event Tracing for Windows (ETW) reader: etwdump

No addresses
No capture filter

Learn



```
No.   Time       Source               Destination   Protocol Lengtl Info
    1 0.000000   CloudNetwork_78:01:… Broadcast     ARP          60 Who has 172.16.72.1? Tell 172.16.76.69
    2 0.542647   CloudNetwork_77:ff:… Broadcast     ARP          60 Who has 172.16.72.1? Tell 172.16.76.55
    3 0.990584   CloudNetwork_78:01:… Broadcast     ARP          60 Who has 172.16.72.1? Tell 172.16.76.69
    4 1.313011   CloudNetwork_77:ff:… Broadcast     ARP          60 Who has 172.16.72.1? Tell 172.16.76.55
    5 1.697712   WNC_73:05:00         Broadcast     ARP          60 ARP Announcement for 172.16.79.143
    6 1.977227   CloudNetwork_78:01:… Broadcast     ARP          60 Who has 172.16.72.1? Tell 172.16.76.69
    7 2.113025   CloudNetwork_78:70:… Broadcast     ARP          60 ARP Announcement for 0.0.0.0
    8 2.122140   XiaomiCommun_54:8b:… Broadcast     ARP          60 ARP Announcement for 0.0.0.0
    9 2.158382   a2:82:4f:71:12:fe    Broadcast     ARP          60 Who has 172.16.72.1? Tell 172.16.79.139
   10 2.313753   CloudNetwork_77:ff:… Broadcast     ARP          60 Who has 172.16.72.1? Tell 172.16.76.55
   11 2.406765   CloudNetwork_78:70:… Broadcast     ARP          60 Who has 172.16.76.102? (ARP Probe)
   12 2.471697   CloudNetwork_77:ff:… Broadcast     ARP          60 Who has 172.16.76.102? Tell 172.16.76.55
   13 2.501244   7a:5c:08:33:48:6d    Broadcast     ARP          60 Who has 172.16.76.102? Tell 172.16.76.92
   14 2.502246   CloudNetwork_78:70:… Broadcast     ARP          60 Who has 172.16.76.102? Tell 172.16.76.97
   15 2.505513   42:b7:30:8e:63:69    Broadcast     ARP          60 Who has 172.16.76.102? Tell 172.16.76.54
   16 2.715569   be:a4:54:ca:2d:8e    Broadcast     ARP          60 Who has 172.16.76.102? Tell 172.16.76.53
   17 2.744630   XiaomiCommun_54:8b:… Broadcast     ARP          60 ARP Announcement for 0.0.0.0
   18 2.977036   CloudNetwork_78:70:… Broadcast     ARP          60 ARP Announcement for 0.0.0.0
   19 2.984878   CloudNetwork_78:01:… Broadcast     ARP          60 Who has 172.16.72.1? Tell 172.16.76.69
   20 3.322185   CloudNetwork_77:ff:… Broadcast     ARP          60 Who has 172.16.76.102? Tell 172.16.76.55
   21 3.423633   CloudNetwork_78:70:… Broadcast     ARP          60 Who has 172.16.76.102? (ARP Probe)
   22 3.512738   CloudNetwork_77:ff:… Broadcast     ARP          60 Who has 172.16.72.1? Tell 172.16.76.55
   23 3.583626   Sophos_cf:be:41      Broadcast     ARP          60 Who has 172.16.76.163? Tell 172.16.72.1
   24 3.605603   82:4b:2b:9d:b8:2a    Broadcast     ARP          60 ARP Announcement for 0.0.0.0
```

```
> Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{5A53B6BE-3E45-4EC6-BB0F-4E980992
> Ethernet II, Src: CloudNetwork_78:01:43 (4c:82:a9:78:01:43), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Address Resolution Protocol (request)
```

```
0000  ff ff ff ff ff ff 4c 82  a9 78 01 43 08 06 00 01   ······L· ·x·C····
0010  08 00 06 04 00 01 4c 82  a9 78 01 43 ac 10 4c 45   ······L· ·x·C··LE
0020  00 00 00 00 00 00 ac 10  48 01 00 00 00 00 00 00   ········ H·······
0030  00 00 00 00 00 00 00 00  00 00 00 00               ········ ····
```